SSM - Mental Health

Temporary Assistance for Needy Families and Children's Mental Health: Evidence from the National Survey of Children's Health (2016-2022)

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Abstract:	Introduction: In the United States, 1 in 5 children live with a mental, behavioral, or developmental disorder. Social safety net programs hold promise as a means through which to improve child well-being. We aimed to evaluate associations between Temporary Assistance for Needy Families (TANF) and four measures of children's mental health: anxiety, depression, attention-deficit and hyperactivity disorder (ADHD/ADD), and behavioral problems. Methods: Our sample included 60,493 children experiencing low income, aged 3-17, who were enrolled in the National Survey of Children's Health (2016-2022). Using linear probability models, we performed both between-state cross-sectional and within-state two-way fixed effects (TWFE) analyses, accounting for time-invariant individual and state policy features and time-variant federal economic trends. Results: In cross-sectional models, a $100 increase in maximum monthly TANF benefits was associated with a lower probability of ADHD/ADD (-0.66 percentage points [pp]; 95% confidence interval [CI] -1.01, -0.32); we observed a marginal association for behavioral problems (-0.41 pp; 95% CI -0.85, 0.03), and non-significant associations for anxiety (-0.11 pp; 95% CI -0.49, 0.27) and depression (-0.04 pp; 95% CI -0.38, 0.30). Cross-sectional associations for ADHD/ADD were more protective in male than female children and were robust across sensitivity analyses that considered symptom severity and alternative lag periods. TWFE results were null. Conclusions: TANF generosity was associated with reduced probabilities of children's externalizing disorders between, but not within, states over time. Future studies should employ quasi-experimental designs over longer periods of time to assess greater variation in benefit generosity and should examine TANF in combination with other

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	state-level social policies.
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Question	**Response**

Cover Letter

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Dr. Alexander Tsai, MD, PhD
Dr. Lesley Jo Weaver, PhD, MPH
Editors-in-Chief
Social Science & Medicine – Mental Health

Dear Dr. Tsai and Dr. Weaver:

Please find attached our manuscript entitled "Temporary Assistance for Needy Families (TANF) and Children's Mental Health: Evidence from the National Survey of Children's Health (2016-2022)" to be considered for publication in *Social Science & Medicine (SSM) – Mental health.*

This paper examines the relationship between state TANF generosity and four measures of mental health —anxiety, depression, attention deficit/hyperactivity disorder (ADHD/ADD), and behavioral problems— in a nationally-representative sample of children, aged 3-17, living in low-income households in the US.

We found increased TANF generosity to be associated with modest reductions in the prevalence of children's externalizing disorders, such as ADHD/ADD and behavioral problems, between but not within states over time; exposure was more protective against diagnosis with ADHD/ADD in male than female children. The prevalence of children's internalizing disorders, including anxiety and depression, was not meaningfully impacted by increased TANF generosity either between or within states over time.

These findings are directly relevant to SSM-Mental Health's audience, as they address an important knowledge gap regarding the influence of social safety net programs on children's emotional well-being.

Our work is theoretically grounded in the Family Stress Model, and we employed two complementary analytic approaches, fitting both cross-sectional and two-way fixed effects models to evaluate associations between and within states over time, respectively. Models were adjusted for a range of individual-, household-, and state-level covariates, and results were robust across sensitivity analyses conducted to assess symptom severity and alternative lag periods. We discuss several areas for future research.

We have identified five potential reviewers for this article based on their relevant expertise. This manuscript has not been previously published. All data leveraged in this paper are publicly available.

Thank you for your consideration.

Sincerely,

Karolina Edlund

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Declaration of Interest Statement:
The authors declare no competing financial or personal interests.

Article Title:
Temporary Assistance for Needy Families and Children's Mental Health: Evidence from the National Survey of Children's Health (2016-2022)

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All data leveraged in this paper are publicly available. Please contact karolinaedlund@hsph.harvard.edu with specific questions.

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ABSTRACT

Introduction: In the United States, 1 in 5 children live with a mental, behavioral, or developmental disorder. Social safety net programs hold promise as a means through which to improve child well-being. We aimed to evaluate associations between Temporary Assistance for Needy Families (TANF) and four measures of children's mental health: anxiety, depression, attention-deficit and hyperactivity disorder (ADHD/ADD), and behavioral problems.

Methods: Our sample included 60,493 children experiencing low income, aged 3-17, who were enrolled in the National Survey of Children's Health (2016-2022). Using linear probability models, we performed both between-state cross-sectional and within-state two-way fixed effects (TWFE) analyses, accounting for time-invariant individual and state policy features and time-variant federal economic trends.

Results: In cross-sectional models, a $100 increase in maximum monthly TANF benefits was associated with a lower probability of ADHD/ADD (-0.66 percentage points [pp]; 95% confidence interval [CI] -1.01, -0.32); we observed a marginal association for behavioral problems (-0.41 pp; 95% CI -0.85, 0.03), and non-significant associations for anxiety (-0.11 pp; 95% CI -0.49, 0.27) and depression (-0.04 pp; 95% CI -0.38, 0.30). Cross-sectional associations for ADHD/ADD were more protective in male than female children and were robust across sensitivity analyses that considered symptom severity and alternative lag periods. TWFE results were null.

Conclusions: TANF generosity was associated with reduced probabilities of children's externalizing disorders between, but not within, states over time. Future studies should employ quasi-experimental designs over longer periods of time to assess greater variation in benefit generosity and should examine TANF in combination with other state-level social policies.

KEY WORDS: children's mental health, social determinants, economic policies

1. INTRODUCTION

Children's mental health remains a critical concern, with an estimated 1 in 5 children in the United States (US) living with a mental, behavioral, or developmental disorder (Perou et al., 2013). Among children aged 3 to 17, the most commonly diagnosed conditions include attention-deficit and hyperactivity disorder (ADHD/ADD; 9.8%), anxiety (9.4%), behavioral problems (8.9%), and depression (4.4%) (Bitsko et al., 2022). Roughly half of all children with mental health diagnoses live without adequate treatment, and more remain undiagnosed altogether (Whitney & Peterson, 2019). COVID-19 further affected children's mental health, resulting in a moderate to two-fold increase in the number of children and adolescents experiencing symptoms during and in the wake of the pandemic (Racine et al., 2021; Sharma et al., 2021). The burdens of mental health disorders also remain disparate. Black or African American and Hispanic children living in poverty face both worse mental health outcomes and less access to care than do their White and more affluent peers (Alegría et al., 2015).

The Family Stress Model (FSM), a theoretical framework designed to explain how household income influences child well-being, provides valuable perspective on the pathways leading to these mental health disparities. As described by Conger et al. in 2002, the FSM posits that insufficient household income increases caregiver psychological distress which, in turn, increases caregiver conflict and affects child and youth development (Conger & Conger, 2002). Economic pressures that force caregiver choices about bills, daily purchases, health insurance, and medical care, not only distress caregivers but also worsen health outcomes for household dependents (Chen et al., 2024; Conger et al., 2010; Landers-Potts et al., 2015; Masarik & Conger, 2017; Neppl et al., 2016). In this framework, social safety net programs that seek to reduce poverty and economic pressure, through income support or other benefits, hold promise as a means through which to improve children's mental health (Gassman-Pines & Hill, 2013).

Prior to President Donald Trump's first term, the US government increased expenditures for social safety net programs (Kogan et al., 2019, May 15; Leachman et al., 2017, July 19). Increased funding varied across the welfare landscape, however, resulting in the expansion of tax credit-based and food assistance programs—including the Earned Income Tax Credit (EITC), Child Tax Credit (CTC), and Supplemental Nutrition Assistance Program (SNAP)—and the contraction of cash assistance-based initiatives such as the Temporary Assistance for Needy Families (TANF) program (Hardy et al., 2018; Hardy, 2017).

The TANF program aims to provide income support to families experiencing economic hardship in the US. Since the program's establishment in 1996 and renewal in 2005, the federal government has supplied $16.5 billion in annual TANF block grants to the fifty states, the District of Columbia (DC), surrounding territories, and tribal lands (Center on Budget and Policy Priorities, 2022). While states are largely responsible for defining their own eligibility criteria, TANF recipients must be US citizens or qualifying non-citizens (e.g., Persons Residing Under Color of Law (PRUCOL)), residents of the state or district or territory, un/underemployed or of low income, have a child younger than 18 years of age, or be pregnant or a head of household younger than 18 years of age, themselves. All state-managed TANF programs must additionally function to address one or more central goals:

> Provide assistance to needy families so that children can be cared for in their own home or in the homes of relatives; End the dependence of needy parents on government benefits by promoting job preparation, work, and marriage; Prevent and reduce the incidence of out-of-wedlock pregnancies; Encourage the formation and maintenance of two-parent families (Office of Family Assistance, 2022).

Over several decades, the federal TANF expenditure has failed to grow in line with inflation, contributing to a roughly 40% reduction in programmatic purchasing power and the lowest ever TANF-to-poverty-level ratio (Schott & Finch, 2010). Substantial inequities also persist in state TANF generosity. For instance, in 2021, an eligible family of three could expect to receive cash benefits equal to just $204 a month (11% of the federal

poverty line and 4% of the state median income) in Arkansas compared to $1,098 a month (60% of the federal poverty line and 12% of the state median income) in New Hampshire (Center on Budget and Policy Priorities, 2022). Such benefit disparities persist at a time when more children than ever report struggling with their mental health, and when an increased number of families could benefit from cash assistance. While pandemic-era stimulus checks and child tax credits contributed to a reduction in US child poverty, the expiration of these economic relief measures saw a reversal of earlier gains, with rates of child poverty more than doubling from 5.2% in 2021 to 12.4% in 2022 (Williams & Rudowitz, 2024).

The relationship between state TANF generosity and improved maternal and child health has been described in the extant literature. For example, Swanson et al. (2020) report modest associations between increased state TANF generosity and reduced infant low birth weight and mortality (Swanson & Chavez, 2020). Additional work has characterized the protective effects of TANF cash benefits against physical child abuse (Spencer et al., 2021). Strict benefit eligibility policies, which limit caseloads, have also been associated with higher neglect victimization and foster care placement (Berger, 2004; Ginther & Johnson-Motoyama, 2022). However, academic consensus has yet to be reached regarding the impact of TANF on emotional well-being. While some studies suggest greater access to TANF resources may reduce maternal depression (Holmes et al., 2020), others emphasize negative or null associations between benefit conditionality and parental psychological distress (Cheng, 2007; Davis, 2019; Spencer et al., 2022; Walker et al., 2023; Wang, 2015).

Our study aimed to address this knowledge gap by evaluating the relationship between state-level maximum monthly TANF cash benefits and four measures of children's mental health—anxiety, depression, ADHD/ADD, and behavioral problems—using two distinct yet complementary analytic approaches. We first employed between-state models to cross-sectionally describe the association between benefit generosity and children's mental health. Next, we fit two-way fixed effects (TWFE) models to explore the within-state dynamics of these associations as states' benefit generosity varied over time, accounting for time-invariant individual and state policy features and time-variant federal economic trends. Models were adjusted for a range of individual and household demographic characteristics, as well as state-level covariates. Consistent with constructs defined in the FSM, we hypothesized that higher benefit generosity would be protective against adverse child mental health outcomes, with differences in effect expected across levels of sex, age, and race and ethnicity.

2. MATERIALS & METHODS

2.1. Study population

The study population included children 3 to 17 years of age who were enrolled in the 2016 to 2022 waves of the National Survey of Children's Health (NSCH), a cross-sectional, mail and web-based questionnaire administered annually by the Health Resources and Services Administration Maternal and Child Health Bureau (Ghandour et al., 2018). As recipients of TANF benefits must be of low-income, we restricted our sample to children residing in households with total pre-taxation incomes less than 200% of the Federal Poverty Level (FPL). From the 279,546 children initially included in the 2016 to 2022 waves of the NSCH, we excluded those who were outside the age range of 3 to 17 (n=40,012), were not residing in low-income households (n=175,041), and were missing any outcome or covariate data (n=4,000). Our analytic sample included 60,493 children (**Figure 1**).

2.2. Measures

2.2.1. Independent Variable

In March of 2024, the first author downloaded TANF policy data from the Urban Institute's Welfare Rules Database (Knowles et al., 2023). Our continuous exposure variable represented the maximum monthly cash benefit provided to a family of three, with no income, residing in each state's most populous area. In such

instances where the reported maximum monthly benefit varied by eligibility group, we included the values affecting the majority of the TANF caseload in each state: California (non-exempt), Massachusetts (exempt through November 29, 2018), and Wisconsin (W-2 Transition). All model estimates were scaled to represent $100 changes in benefits and lagged by one year, to account for the policy's potential delayed effects on child health (Spencer et al., 2021).

2.2.2. Outcome Variables

In primary analyses, we examined four dichotomous measures of children's mental health, including anxiety (yes or no), depression (yes or no), ADHD/ADD (yes or no), and behavioral problems (yes or no). Each outcome was deemed clinician-diagnosed and current at the time of surveying based on caregiver responses to two sequential NSCH questions: (1) Has a doctor or other health care provider ever told you that this child has the [mental health condition]? and (2) If yes, does this child currently have the [mental health condition]? Educators such as teachers and school nurses were additionally included in the first question stem for behavioral problems. Consistent with the NSCH literature, children whose caregivers had ever been told that they had a condition, but who were not currently experiencing symptoms, were not classified as having that outcome (Kavanagh et al., 2024).

In sensitivity analyses, we further examined the influence of symptom severity for each outcome (no condition/mild symptoms or moderate/severe symptoms) based on caregiver responses to a third NSCH question: (3) Would you describe this child's current [mental health condition] as mild, moderate, or severe?

2.2.3. Individual-level Control Variables

Using demographic data provided by caregivers to the NSCH, we controlled for each child's reported age (years), sex (male or female), combined race and ethnicity (White; Hispanic of any race; Black of African American; Asian, Native Hawaiian, or Pacific Islander; American Indian or Alaska Native; or Other or mixed race), family nativity (child and parent born in the US; child and/or parent not born in the US; or child born in the US with missing parent data), family structure (two married parents; two unmarried parents; single parent; or other), and highest household adult's educational attainment (college degree or higher; some college or associate degree; high school (including vocational); or less than high school).

2.2.4. State-level Control Variables

Because TANF is not the sole time-varying state policy likely to affect families experiencing low income, our fully-adjusted models additionally controlled for each state's minimum wage (US dollars), Medicaid income eligibility limits for children aged 1-5 years (percent of the FLP), Medicaid income eligibility limits for children aged 6-18 years (percent of the FLP), and Earned Income Tax Credit (EITC) status (refundable EITC; non-refundable EITC; or no EITC). In early 2024, minimum wage, Medicaid, and EITC data were sourced from the Bureau of Labor Statistics, Kaiser Family Foundation, and University of Kentucky's Center for Poverty Research National Welfare Database, respectively (University of Kentucky Center for Poverty Research, 2023).

2.3. Statistical Analysis

We fitted separate linear probability models using ordinary least squares regression to examine between-state cross-sectional associations between one-year-lagged maximum monthly TANF cash benefits (in continuous $100 units) and each mental health outcome. We next explored the within-state dynamics of these associations using separate ordinary least squares, TWFE models, which included state fixed effects to address time-invariant individual and policy features and year fixed effects for time-variant federal economic trends. All models incorporated survey weights and clustered standard errors at the state level and were estimated both with and without controls; Model 1 was unadjusted, Model 2 adjusted for individual- and household-level covariates, and Model 3 additionally adjusted for state-level covariates. Hereafter, Model 3 is referred to as fully-adjusted.

Between-state cross-sectional models evaluate whether states with higher benefit generosity have more favorable probabilities of children's mental health diagnoses. Yet, many state-level characteristics beyond TANF—such as other economic policies, culture, and contextual factors—could explain observed associations. Within-state TWFE models account for several additional classes of potential confounders, including fixed state-level characteristics and national trends; as such, they likely present more robust estimates of how changing a state's TANF cash benefit generosity impacts children's mental health.

2.3.1. Sensitivity Analyses

Mental health symptom patterns are known to differ by sex, age, and race and ethnicity. Specifically, female children have been found to exhibit higher levels of internalizing symptoms, while male children exhibit higher levels of externalizing symptoms (Haahr-Pedersen et al., 2021). Age is also positively associated with internalizing pathology, and rates of adverse mental health outcomes remain highest among racial minorities compared to White children (Alegría et al., 2015; Stewart et al., 2022). Most families that receive TANF include at least one child under the age of 12 (US Department of Health & Human Services, 2023). Child TANF recipients are also diverse, with 25% identifying as White, 28% as Black, and 39% as Hispanic of any race in 2022 (US Department of Health & Human Services, 2023).

Our sensitivity analyses therefore explored associations (1) by sex groups, (2) by age groups, (3) by race and ethnicity groups, (4) using alternative forms of each outcome to consider both diagnostic status and symptom severity, and (5) using contemporaneous and two-year-lagged models. We assessed potential effect measure modification by sex, age, and race and ethnicity using interaction terms and Wald tests. Between-state cross-sectional and within-state TWFE models, as described above, were employed throughout.

3. RESULTS

Our analytic sample included 60,493 children experiencing low income, aged 3 to 17, with complete outcome and covariate data per caregiver reports to the 2016 to 2022 waves of the NSCH (**Figure 1**). In this weighted sample, the mean age was 10.01 years old (standard deviation (SD) = 4.29) and 52% were male. Approximately 33%, 37%, and 20% of participants identified as White, Hispanic of any race, and Black or African American, respectively. The majority of children and their parents were born in the US (56%), and many resided in households with two married parents (40%). Caregiver educational attainment varied, with 61% of caregivers having completed between a high school degree and some college. ADHD/ADD (10%) and behavioral problems (9%) were the most commonly diagnosed conditions, followed by anxiety (8%), and depression (4%, **Table 1**).

Maximum monthly TANF cash benefits varied between states and increased in 27 states and DC across time, rising from an average of $442 (SD = $167) in 2015 to $501 (SD = $201) in 2021 (**Figures 2-4, Table A1**). The mean change in cash benefits between consecutive years was $9.75 (2%) and between 2015 and 2021 was $58.50 (14%; **Table A1**).

3.1. Between-State Cross-Sectional Findings

In fully-adjusted models, a $100 increase in maximum monthly TANF cash benefits was associated with a lower probability of ADHD/ADD (-0.66 percentage points [pp]; 95% confidence interval [CI] -1.01, -0.32). We observed a marginal association for behavioral problems (-0.41 pp; 95% CI -0.85, 0.03), and non-significant associations for anxiety (-0.11 pp; 95% CI -0.49, 0.27) and depression (-0.04 pp; 95% CI -0.38, 0.30; **Table 2**).

No meaningful differences in the fully-adjusted probabilities of ADHD/ADD, behavioral programs, anxiety, or depression were observed between groups of race and ethnicity (**Table A2**). Increased cash benefits did, however, appear more protective against ADHD/ADD in male children (-0.96 pp; 95% CI -1.52, -0.40) than female children (-0.37 pp; 95% CI -0.62, -0.12; Wald p 0.0229), as well as protective against anxiety in children aged 3 to 11 (-

0.32 pp; 95% CI -0.62, -0.02) but not 12 to 17 (0.20 pp; 95% CI -0.40, 0.80; Wald p 0.0276). Probabilities of anxiety, depression, and behavioral problems did not significantly vary by sex group, neither did probabilities of depression, ADHD/ADD, and behavioral problems by age group.

Estimates obtained when exploring the impact of exposure on moderate to severe forms of each outcome and when using contemporaneous and two-year lagged models were consistent with our primary results (**Tables A3 and A4**).

3.2. Within-State TWFE Findings

Fully-adjusted estimates were null for ADHD/ADD (-0.83 pp; 95% CI -2.02, 0.37), behavioral problems (2.51 pp; 95% CI -0.49, 5.51), anxiety (1.22 pp; 95% CI -0.61, 3.04), and depression (1.47 pp; 95% CI -0.41, 3.36) for all outcome variations and lag periods (**Tables 2, A3 and A4**).

Although fully-adjusted estimates were non-significant, increased cash benefits again appeared more protective against ADHD/ADD in male children (-1.12 pp; 95% CI -3.29, 1.06) than female children (-0.54 pp; 95% CI -1.56, 0.47; Wald p 0.0317), and protective against anxiety in children aged 3 to 11 (-0.21 pp; 95% CI -1.56, 1.13) but not 12 to 17 (3.26 pp; 95% CI 0.37, 6.15; Wald p 0.0272; **Table A2**). As before, we did not observe meaningful effect measure modification for any outcome by race and ethnicity.

4. DISCUSSION

In a nationally representative sample of children experiencing low income, a $100 increase in maximum monthly TANF cash benefits reduced the probabilities of externalizing disorders such as ADHD/ADD and, to a marginal degree, behavioral problems. We observed null associations for internalizing disorders such as anxiety and depression. TANF generosity was found to be more protective for male than female children with respect to ADHD/ADD, and protective for younger but not older children with respect to anxiety. These between-state cross-sectional results were robust across sensitivity analyses conducted using alternative forms of each outcome to account for both diagnosis and symptom severity as well as alternative lag periods. While these findings suggest that TANF generosity may be associated with children's mental health, the results of our TWFE models were null, indicating that within-state changes in cash benefits did not meaningfully improve our outcomes.

We can consider several potential explanations for these observed differences between modeling approaches. Decisions regarding the allocation of TANF funds are largely decentralized through the provision of federal block grants, to which states are required to contribute. According to Samudra et al. (2022), on average, states allocate just over one-third of TANF funds to cash assistance, just over one-third to work support, and smaller proportions to other categories such as family formation activities (Fusaro, 2021). However, TANF policy environments, expenditures, and eligibility requirements vary substantially by state. For example, states with liberal electoral politics tend to prioritize greater welfare spending than states under conservative control. Recent changes in state employment rates have also been associated with the provision of more basic than alternative assistance (Fusaro, 2021). While the majority of states choose to cap lifetime TANF receipt at the federal limit of 60 months or beyond, others adopt more stringent limits ranging from 12 to 24 months. States also vary with regard to sanctions following employment noncompliance, with most terminating all benefits or enrollment and others imposing only partial penalties (Knowles, 2023).

While initially distributed in 2003, the NSCH was not annually administered until 2016. The availability of consecutive years of NSCH data thus limited the number of years across which we could observe state-level TANF generosity, as well as the range of measured variation in cash benefit amounts. The average within-state change in maximum monthly benefits from 2015 to 2021 was just $58.50 (14%), and limited exposure variation within states over time may have hindered our ability to detect the effects of change in our TWFE analyses.

Another potential explanation for our findings is the decline in TANF's reach over time. In 1996, approximately 68 out of every 100 families with children in poverty received cash assistance. In 2020, this number had fallen to just 21, resulting in the lowest ever TANF-to-poverty ratio. In 2020 alone, it is estimated that 2.38 million eligible families remained without TANF support (Shrivastava & Thompson, 2022). Eligible families face several barriers to receiving TANF assistance. For instance, the value of federally provided block grants has failed to grow in line with inflation resulting in insufficient state funding and reduced participation (Schott & Finch, 2010). Families may also be unaware of their eligibility status or deterred from applying for coverage by welfare-related stigma or administrative burdens (Stuber & Kronebusch, 2004). Sanctions also threaten continued TANF enrollment. Herein lies an ecological fallacy—increases in state-level TANF cash benefits may not be reaching those most in need.

Our mixed findings across modeling approaches could also be due to unmeasured confounding by coexisting welfare programs. In contrast to TANF, inflation-adjusted spending on programs such SNAP has substantially increased in recent decades, rising by 290% from 2000 to 2012 (Hardy et al., 2018). The COVID-19 pandemic further transformed the US welfare landscape as, under The American Rescue Plan Act, millions of Americans received thousands of dollars in stimulus payments and Child Tax Credit. During this time, no federal policy changes were made to TANF, although states maintained flexibility to adjust eligibility requirements (Shantz et al., 2023). It follows that the effects of comparatively modest TANF cash benefits may be overshadowed by those of more generous programs.

The patterns of association we observed between TANF generosity and externalizing disorders reflect those described in the literature (Magnuson & Votruba-Drzal, 2009; Miller & Votruba-Drzal, 2017). Using data from The Great Smoky Mountains Study and a marginal model approach, Costello et al. (2003) reported significant reductions in conduct and oppositional defiant disorder symptoms among American Indian children exposed to increased household income following the establishment of a local gambling casino. In contrast, associations for symptoms of anxiety and depression remained largely unchanged (Costello et al., 2003). Similarly, Dearing et al. (2006) found higher household income to be negatively linked to within-child estimates of externalizing problems in the Study of Early Child Care and Youth Development, although within-child estimates for internalizing problems were nonsignificant (Dearing et al., 2006). Such findings are further supported by Woods-Jaeger et al. (2021), who, using data from the Future of Families and Child Wellbeing Study and a difference-in-difference approach, demonstrated reductions in externalizing behaviors following an increase in minimum wage (Woods-Jaeger et al., 2021). Finally, in a systematic review, Russell et al. (2015) identified consistent ties between low socioeconomic status and heightened risk of childhood ADHD across 42 studies (Russell et al., 2016). Children and adolescents exposed to adversity, such as poverty, are thought to be more likely to "act out" than internalize their emotions due to limited self-control stemming from nascent prefrontal cortex development (Miller & Cohen, 2001).

If changes in household income are more protective against child externalizing than internalizing symptoms, and male children are more likely than female children to be diagnosed with externalizing disorders, it is perhaps unsurprising that we observed effect measure modification of the relationship between TANF and ADHD/ADD by child sex (Haahr-Pedersen et al., 2021). Early childhood also represents a sensitive developmental period marked by heightened reliance on caregivers and the home environment. Our finding that TANF generosity is protective against anxiety only in earlier childhood is thus in line with FSM hypotheses (Fields et al., 2021).

Like Spencer et al. (2021), who observed no difference in the effect of TANF on child maltreatment across White and Black or African American mothers, we observed no difference in the impact of TANF on mental health across White and non-White children (Spencer et al., 2021). While TANF cash benefits do not appear to directly contribute to racial disparities in children's mental health, the racist legacy of welfare allocation in the US is well-described (Floyd et al., 2021, August 4; Parolin, 2021; Rodgers Jr & Tedin, 2006). States with greater racial and

ethnic caseload diversity are known to favor alternative over traditional cash transfers, and non-White families are more likely to live in states with strict work requirements, time limits, and low monthly benefits (Parolin, 2021). In this context, the racially disparate allocation of TANF cash benefits perpetuates structural racism by contributing to cycles of poverty that shape children's health (Monnat, 2010). Our findings would benefit from replication in a sample that does not necessitate the dichotomization of race and ethnicity.

4.1. Strengths & Limitations

Our findings bridge an important knowledge gap regarding the influence of TANF on emotional well-being, especially among children. Our work is theoretically grounded in the FSM, and we employed two complementary analytic approaches, fitting both cross-sectional and TWFE models to evaluate associations between and within states over time, respectively. We assessed four mental health outcomes in a nationally-representative sample of children experiencing low income, controlling for a range of individual-, household- and state-level covariates. Though modest, our population-level findings contribute to a growing evidence base in a time of declining investments in cash assistance-based initiatives such as TANF.

Our results must, however, be interpreted with respect to several limitations. First, rather than matching families to the exact benefit amounts for which they were eligible, we used a state-level measure representing the maximum monthly cash benefits available to a family of three, with no additional income, living in each state's most populous area. Despite the fact that several states' benefits vary by program eligibility status, region, or county, we consistently used the benefit amount affecting the majority of each state's TANF caseload. This approach has been previously described by Spencer et al. (2021), although it may lead to imprecise benefit amounts at the individual level and bias towards the null (Spencer et al., 2021). Second, we obtained child mental health data via caregiver report, and only reported conditions were available; for example, the NSCH does not include validated emotional or behavioral symptom scales. A related concern is that diagnosis is a "strict" outcome. It is possible that many children did see an improvement in their mental health that failed to rise to the level of a diagnosis. That said, sensitivity models examining the severity of diagnoses did not yield different results. Finally, estimates obtained using within-state TWFE models are susceptible to bias arising from staggered policy implementation during the study period. Evaluation of the impact of additional, time-variant policy characteristics—including diversion programs, benefit time limits, work sanctions, and family caps—was beyond the scope of the present study but represents an important area for future research (Callaway & Sant'Anna, 2021; Goodman-Bacon, 2021).

5. CONCLUSIONS

This paper provides an in-depth examination of the relationship between TANF generosity and four measures of children's mental health, using two different analytic approaches. While our between-state cross-sectional findings suggest more generous cash benefits may be associated with lower probabilities of children's externalizing disorders such as ADHD/ADD and, albeit marginally, behavioral problems, our TWFE models were null, failing to establish a link between within-state changes in TANF generosity and children's mental health over time. Given this inconclusive evidence, further research is needed. Studies employing quasi-experimental designs with larger variations in TANF generosity over time could provide more robust insights. Additionally, examining the role of other state-level policies in combination with TANF could provide a more holistic picture of how economic safety nets impact children's health (Matthay et al., 2021).

6. ACKNOWLEDGEMENTS

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7. CRediT AUTHOR CONTRIBUTIONS

Karolina M. Edlund: Conceptualization, Investigation, Methodology, Data Curation, Software, Formal Analysis, Visualization, Writing-Original Draft; **Nolan M. Kavanagh:** Data Curation; Software, Formal Analysis, Visualization, Writing-Review & Editing; **Vinícius Prado:** Conceptualization, Investigation, Visualization, Writing-Review & Editing; **Rachael A. Lennon:** Resources, Writing-Review & Editing; **Natalie Slopen:** Supervision, Writing-Reviewing and Editing

REFERENCES

Alegría, M., Green, J. G., McLaughlin, K. A., & Loder, S. (2015). Disparities in child and adolescent mental health and mental health services in the US. *New York, NY: William T. Grant Foundation*, 26.

Berger, L. M. (2004). Income, family structure, and child maltreatment risk. *Children and Youth Services Review*, *26*(8), 725-748.

Bitsko, R., Claussen, A., Lichtstein, J., Black, L., Everett Jones, S., Danielson, M., Hoenig, J., Davis Jack, S., Brody, D., & Gyawali, S. (2022). Surveillance of children's mental health–United States, 2013–2019. *MMWR*, *71*(2), 1-42.

Callaway, B., & Sant'Anna, P. H. (2021). Difference-in-differences with multiple time periods. *Journal of econometrics*, *225*(2), 200-230.

Center on Budget and Policy Priorities. (2022, March 1). *Policy Basics: Temporary Assistance for Needy Families*. https://www.cbpp.org/research/family-income-support/policy-basics-an-introduction-to-tanf

Chen, J.-H., Huang, C.-H., Wu, C.-F., Jonson-Reid, M., & Drake, B. (2024). The application of family stress Model to Investigating adolescent problematic behaviors: The moderating role of assets. *Journal of Family and Economic Issues*, *45*(1), 174-183.

Cheng, T. C. (2007). Impact of work requirements on the psychological well-being of TANF recipients. *Health & Social Work*, *32*(1), 41-48.

Conger, R. D., & Conger, K. J. (2002). Resilience in Midwestern families: Selected findings from the first decade of a prospective, longitudinal study. *Journal of marriage and family*, *64*(2), 361-373.

Conger, R. D., Conger, K. J., & Martin, M. J. (2010). Socioeconomic status, family processes, and individual development. *Journal of marriage and family*, *72*(3), 685-704.

Costello, E. J., Compton, S. N., Keeler, G., & Angold, A. (2003). Relationships between poverty and psychopathology: A natural experiment. *Jama*, *290*(15), 2023-2029.

Davis, O. (2019). What is the relationship between benefit conditionality and mental health? Evidence from the United States on TANF policies. *Journal of Social Policy*, *48*(2), 249-269.

Dearing, E., McCartney, K., & Taylor, B. A. (2006). Within-child associations between family income and externalizing and internalizing problems. *Developmental psychology*, *42*(2), 237.

Fields, A., Harmon, C., Lee, Z., Louie, J. Y., & Tottenham, N. (2021). Parent's anxiety links household stress and young children's behavioral dysregulation. *Developmental psychobiology*, *63*(1), 16-30.

Floyd, I., Pavetti, L., Meyer, L., Safawi, A., Schott, L., Bellew, E., & Magnus, A. (2021, August 4). TANF policies reflect racist legacy of cash assistance. *Center on Budget and Policy Priorities*. https://www.cbpp.org/research/income-security/tanf-policies-reflect-racist-legacy-of-cash-assistance

Fusaro, V. A. (2021). State variation in TANF expenditures: Implications for social work and social policy. *Social work*, *66*(2), 157-166.

Gassman-Pines, A., & Hill, Z. (2013). How social safety net programs affect family economic well-being, family functioning, and children's development. *Child Development Perspectives*, *7*(3), 172-181.

Ghandour, R. M., Jones, J. R., Lebrun-Harris, L. A., Minnaert, J., Blumberg, S. J., Fields, J., Bethell, C., & Kogan, M. D. (2018). The design and implementation of the 2016 National Survey of Children's Health. *Maternal and child health journal*, *22*, 1093-1102.

Ginther, D. K., & Johnson-Motoyama, M. (2022). Associations Between State TANF Policies, Child Protective Services Involvement, And Foster Care Placement: Study examines associations between state-level Temporary Assistance for Needy Families policies, child protective services involvement, and foster care placement. *Health Affairs*, *41*(12), 1744-1753.

Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of econometrics*, *225*(2), 254-277.

Haahr-Pedersen, I., Hyland, P., Hansen, M., Perera, C., Spitz, P., Bramsen, R. H., & Vallières, F. (2021). Patterns of childhood adversity and their associations with internalizing and externalizing problems among at-risk boys and girls. *Child Abuse & Neglect*, *121*, 105272.

Hardy, B., Smeeding, T., & Ziliak, J. P. (2018). The changing safety net for low-income parents and their children: Structural or cyclical changes in income support policy? *Demography*, *55*(1), 189-221.

Hardy, B. L. (2017). Income instability and the response of the safety net. *Contemporary Economic Policy*, *35*(2), 312-330.

Holmes, S. C., Ciarleglio, M. M., Song, X., Clayton, A., & Smith, M. V. (2020). Testing the family stress model among Black women receiving Temporary Assistance for Needy Families (TANF). *Journal of Child and Family Studies*, *29*(10), 2667-2677.

Kavanagh, N. M., McConnell, M., & Slopen, N. (2024). State Minimum Wage and Mental Health Among Children and Adolescents. *JAMA Network Open*, *7*(10), e2440810.

Knowles, S., Dehry, I., Shantz, K., & Giannarelli, L. (2023). Welfare Rules Databook: State TANF Policies as of July 2021.

Knowles, S., Dehry, I., Shantz, K., & Goldsmith, L. (2023). *Graphical overview of state and territory TANF policies as of July 2022 (OPRE Report 2023-328)*. https://www.urban.org/sites/default/files/2024-02/Graphical%20Overview%20of%20State%20and%20Territory%20TANF%20Policies%20as%20of%20July%202022.pdf

Kogan, R., Katch, H., Rosenbaum, D., Rice, D., Romig, K., Floyd, I., & Parrott, S. (2019, May 15). Cuts to Low-Income Assistance Programs in President Trump's 2020 Budget Are Wide-Ranging. *Center on Budget and Policy Priorities*. https://www.cbpp.org/research/federal-budget/cuts-to-low-income-assistance-programs-in-president-trumps-2020-budget-are

Landers-Potts, M. A., Wickrama, K., Simons, L. G., Cutrona, C., Gibbons, F. X., Simons, R. L., & Conger, R. (2015). An extension and moderational analysis of the family stress model focusing on African American adolescents. *Family Relations*, *64*(2), 233-248.

Leachman, M., Rosenbaum, D., & Wolkomir, E. (2017, July 19). President Trump's Budget Would Shift SNAP Costs to States, Increasing Risk of Hunger and Weakening Response to Recessions. *Center on Budget and Policy Priorities*. https://www.cbpp.org/research/president-trumps-budget-would-shift-snap-costs-to-states-increasing-risk-of-hunger-and

Magnuson, K., & Votruba-Drzal, E. (2009). Enduring influences of childhood poverty. *Changing poverty, changing policies*, 153-179.

Masarik, A. S., & Conger, R. D. (2017). Stress and child development: A review of the Family Stress Model. *Current opinion in psychology*, *13*, 85-90.

Matthay, E. C., Gottlieb, L. M., Rehkopf, D., Tan, M. L., Vlahov, D., & Glymour, M. M. (2021). What to do when everything happens at once: analytic approaches to estimate the health effects of co-occurring social policies. *Epidemiologic reviews*, *43*(1), 33-47.

Miller, E. K., & Cohen, J. D. (2001). An integrative theory of prefrontal cortex function. *Annual review of neuroscience*, *24*(1), 167-202.

Miller, P., & Votruba-Drzal, E. (2017). The role of family income dynamics in predicting trajectories of internalizing and externalizing problems. *Journal of abnormal child psychology*, *45*, 543-556.

Monnat, S. M. (2010). The color of welfare sanctioning: Exploring the individual and contextual roles of race on TANF case closures and benefit reductions. *The Sociological Quarterly*, *51*(4), 678-707.

Neppl, T. K., Senia, J. M., & Donnellan, M. B. (2016). Effects of economic hardship: Testing the family stress model over time. *Journal of family psychology*, *30*(1), 12.

Office of Family Assistance. (2022, June 29). *About TANF*. Retrieved July 20, 2024 from https://www.acf.hhs.gov/ofa/programs/tanf/

Parolin, Z. (2021). Temporary Assistance for Needy Families and the Black–White child poverty gap in the United States. *Socio-Economic Review*, *19*(3), 1005-1035.

Perou, R., Bitsko, R. H., Blumberg, S. J., Pastor, P., Ghandour, R. M., Gfroerer, J. C., Hedden, S. L., Crosby, A. E., Visser, S. N., & Schieve, L. A. (2013). Mental health surveillance among children—United States, 2005–2011. *MMWR suppl*, *62*(2), 1-35.

Racine, N., McArthur, B. A., Cooke, J. E., Eirich, R., Zhu, J., & Madigan, S. (2021). Global prevalence of depressive and anxiety symptoms in children and adolescents during COVID-19: a meta-analysis. *JAMA pediatrics*, *175*(11), 1142-1150.

Rodgers Jr, H. R., & Tedin, K. L. (2006). State TANF spending: Predictors of state tax effort to support welfare reform. *Review of Policy Research*, *23*(3), 745-759.

Russell, A. E., Ford, T., Williams, R., & Russell, G. (2016). The association between socioeconomic disadvantage and attention deficit/hyperactivity disorder (ADHD): a systematic review. *Child Psychiatry & Human Development*, *47*, 440-458.

Schott, L., & Finch, I. (2010). TANF benefits are low and have not kept pace with inflation. *Center on Budget and Policy Priorities*.

Shantz, K., Knowles, S., Dehry, I., & Giannarelli, L. (2023). State TANF Policies During the COVID-19 Pandemic.

Sharma, M., Idele, P., Manzini, A., Aladro, C., Ipince, A., Olsson, G., Banati, P., & Anthony, D. (2021). Life in Lockdown: Child and Adolescent Mental Health and Well-Being in the Time of COVID-19. *UNICEF Office of Research-Innocenti*.

Shrivastava, A., & Thompson, G. A. (2022). TANF cash assistance should reach millions more families to lessen hardship. *Center on Budget and Policy Priorities*. https://www.cbpp.org/research/income-security/tanf-cash-assistance-should-reach-millions-more-families-to-lessen

Spencer, R. A., Lemon, E. D., Komro, K. A., Livingston, M. D., & Woods-Jaeger, B. (2022). Women's lived experiences with temporary assistance for needy families (TANF): how TANF can better support Women's wellbeing and reduce intimate partner violence. *International journal of environmental research and public health*, *19*(3), 1170.

Spencer, R. A., Livingston, M. D., Komro, K. A., Sroczynski, N., Rentmeester, S. T., & Woods-Jaeger, B. (2021). Association between Temporary Assistance for Needy Families (TANF) and child maltreatment among a cohort of fragile families. *Child Abuse & Neglect*, *120*, 105186.

Stewart, S. L., Lapshina, N., Semovski, V., & Usova, A. (2022). Age, Sex and Relationship Strengths: Internalizing Symptom Differences in Children and Youth Within a Clinical Sample. *Journal of the Canadian Academy of Child and Adolescent Psychiatry*, *31*(4), 189.

Stuber, J., & Kronebusch, K. (2004). Stigma and other determinants of participation in TANF and Medicaid. *Journal of Policy Analysis and Management*, *23*(3), 509-530.

Swanson, J., & Chavez, J. (2020). An evaluation of social assistance programs on infant health. *Social Work and Social Welfare*, *2*, 53-62.

University of Kentucky Center for Poverty Research. (2023). *UKCPR National Welfare Data, 1980–2021*. Retrieved February 18, 2024 from https://ukcpr.uky.edu/resources/national-welfare-data

US Department of Health & Human Services. (2023, October 30). *Characteristics and Financial Circumstances of TANF Recipients, Fiscal Year 2022*. https://www.acf.hhs.gov/sites/default/files/documents/ofa/fy2022_characteristics.pdf

Walker, A., Spencer, R. A., Lemon, E., Woods-Jaeger, B., Komro, K. A., & Livingston, M. D. (2023). The impact of temporary assistance for needy families benefit requirements and sanctions on maternal material hardship, mental health, and parental aggravation. *Maternal and child health journal*, *27*(8), 1392-1400.

Wang, J. S.-H. (2015). TANF coverage, state TANF requirement stringencies, and child well-being. *Children and Youth Services Review*, *53*, 121-129.

Whitney, D. G., & Peterson, M. D. (2019). US national and state-level prevalence of mental health disorders and disparities of mental health care use in children. *JAMA pediatrics*, *173*(4), 389-391.

Williams, E., & Rudowitz, R. (2024). *Recent Trends in Children's Poverty and Health Insurance as Pandemic-Era Programs Expire*. https://www.kff.org/medicaid/recent-trends-in-childrens-poverty-and-health-insurance-as-pandemic-era-programs-expire/

Woods-Jaeger, B., Livingston, M. D., Lemon, E. D., Spencer, R. A., & Komro, K. A. (2021). The effect of increased minimum wage on child externalizing behaviors. *Preventive medicine reports*, *24*, 101627.

Figure 1
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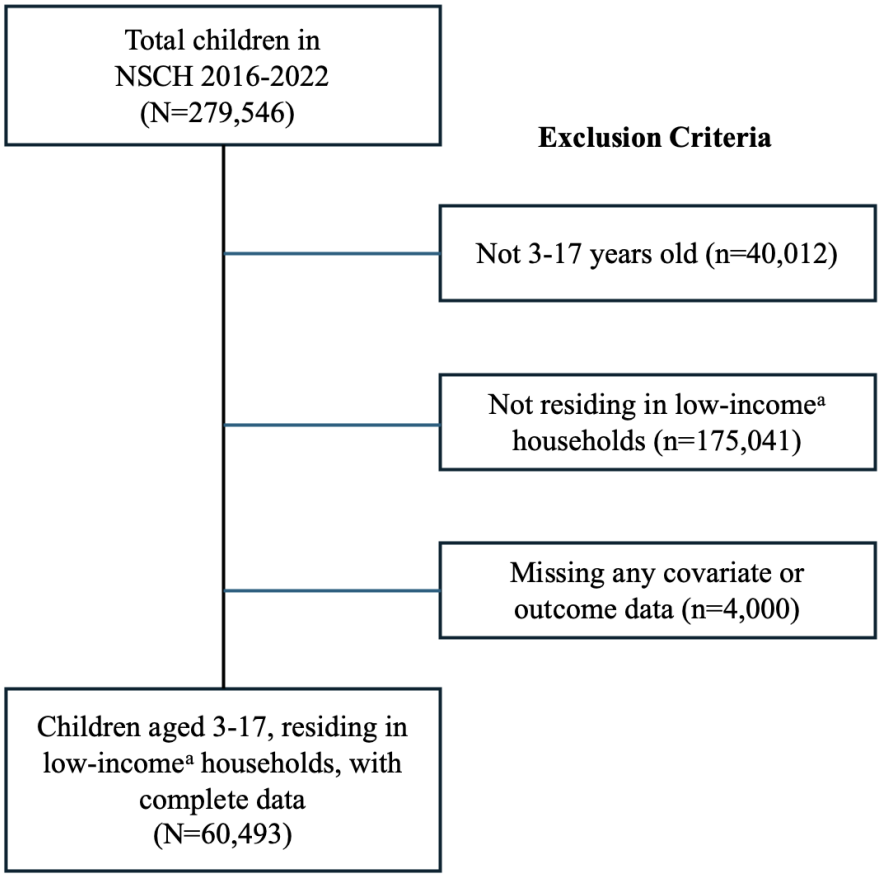


Figure 1. Flowchart illustrating the inclusion and exclusion criteria for our analytic sample (N=60,493).
[a] Low-income defined as <200% of the Federal Poverty Level.

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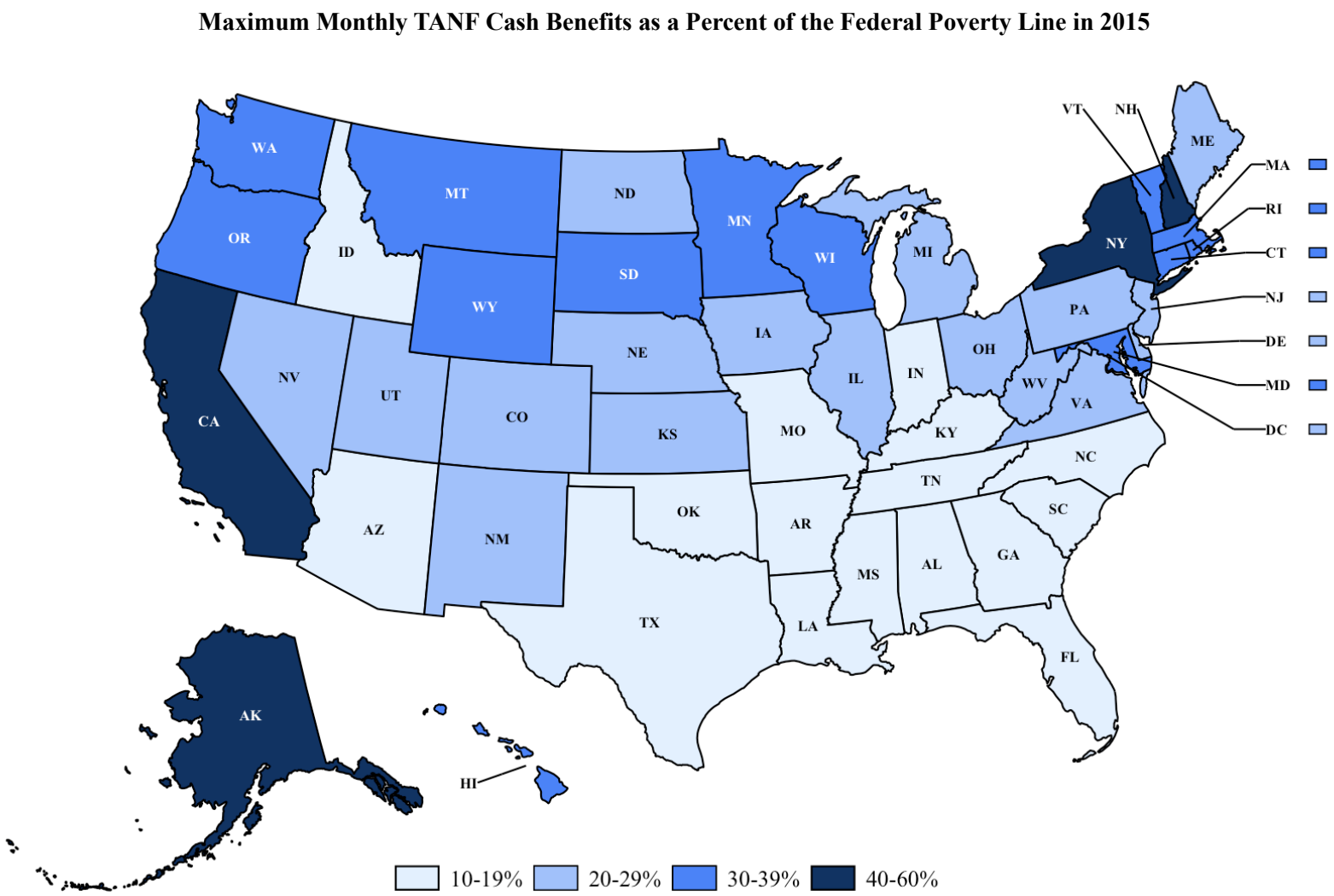


Figure 2. Maximum monthly TANF cash benefits for a family of three with no additional income living in each state's most populous area as a percent of the federal poverty line for a family of three in 2015. Per the 2015 Health and Human Services Poverty Guidelines, the federal poverty line was $1,674/month in the 48 contiguous states and the District of Columbia, $2,093/month in Alaska, and $1,925/month in Hawaii. Figure design adapted from the Center on Budget and Policy Priorities.

Figure 3

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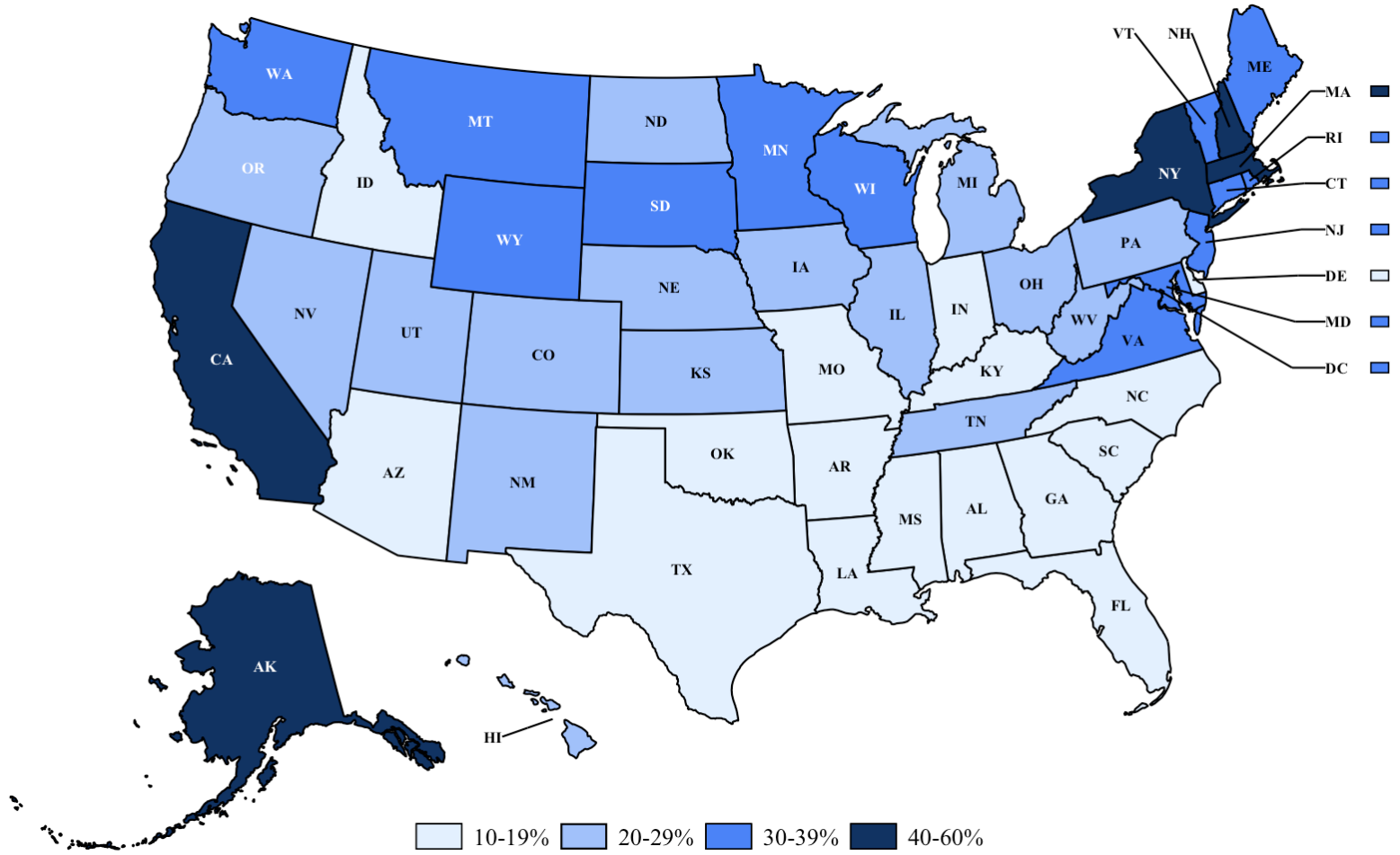


Figure 3. Maximum monthly TANF cash benefits for a family of three with no additional income living in each state's most populous area as a percent of the federal poverty line for a family of three in 2021. Per the 2021 Health and Human Services Poverty Guidelines, the federal poverty line was $1,830/month in the 48 contiguous states and the District of Columbia, $2,288/month in Alaska, and $2,105/month in Hawaii. Figure design adapted from the Center on Budget and Policy Priorities.

Figure 4

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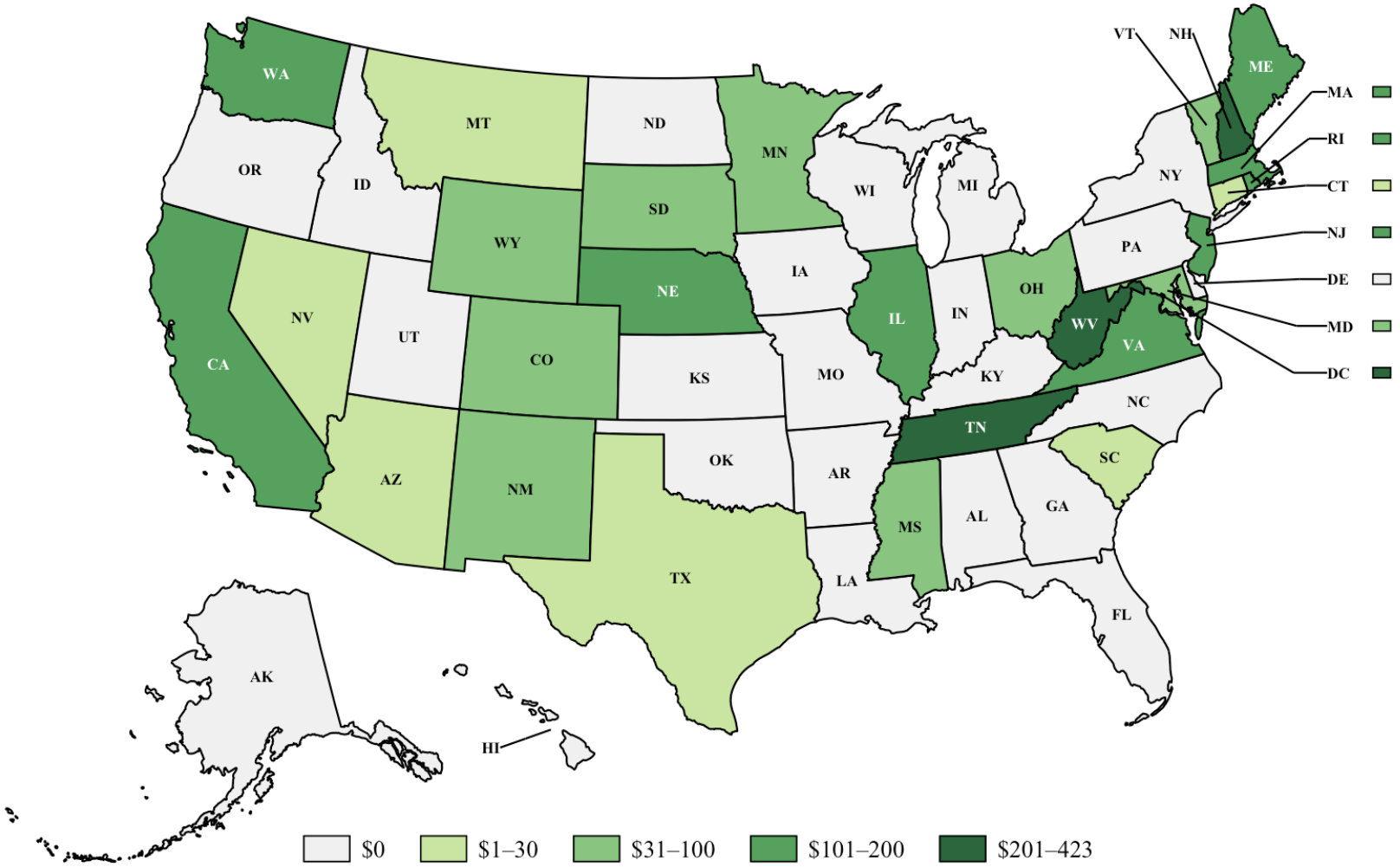


Figure 4. Changes in maximum monthly TANF cash benefits for a family of three with no additional income living in each state's most populous area between 2015 and 2021. Figure design adapted from the Center on Budget and Policy Priorities.

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Table 1. Characteristics of children aged 3-17, residing in low-income[a] households, with complete outcome and covariate data per caregiver report to the National Survey of Children's Health (2016-2022).

Characteristic	**Unweighted** N=60,493	%	**Weighted**
Age			
Mean, years (SD)[b]	10.20 (4.45)	/	10.01 (4.29)
Range	3-17	/	3-17
Sex			
Male	31539	52%	52%
Female	28954	48%	48%
Race and ethnicity			
White[c]	31474	52%	33%
Hispanic of any race	13145	22%	37%
Black or African American[c]	7626	13%	20%
Asian, Native Hawaiian, or Pacific Islander[c]	3304	5%	4%
American Indian or Alaska Native[c]	644	1%	1%
Other or mixed race[c]	4300	7%	5%
Nativity			
Child and parent born in US	41678	69%	56%
Child and or parent born out of US	13569	22%	34%
Child born in US, parent data missing	5246	9%	10%
Family structure			
Two parents, married	25269	42%	40%
Two parents, unmarried	4588	8%	9%
Single parent	4958	8%	8%
Other	17760	29%	29%
Highest adult education			
College degree or higher	17433	29%	20%
Some college or associate degree	20940	35%	27%
High school (including vocational)	17541	29%	34%
Less than high school	4579	8%	19%
Anxiety diagnosis			
Yes	7295	12%	8%
No	53198	88%	92%
Depression diagnosis			
Yes	3787	6%	4%
No	56706	94%	96%
ADHD/ADD[d] diagnosis			
Yes	7496	12%	10%
No	52997	88%	90%
Behavioral problems diagnosis			
Yes	6443	11%	9%
No	54050	89%	91%

[a] Low-income defined as <200% of the Federal Poverty Level.
[b] Standard deviation (SD).
[c] Non-Hispanic.
[d] Attention-deficit and hyperactivity disorder (ADHD/ADD).

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Table 2. Multivariable-adjusted[a] associations between a $100 increase in maximum cash benefits lagged by 1 year and the percentage-point prevalence of each mental health outcome among children aged 3-17, residing in low-income[b] households, with complete covariate and outcome data (N=60,493). Results were obtained via cross-sectional and two-way fixed effects (TWFE) models, with standard errors clustered at the state level.

		Cross-sectional			TWFE[c]		
		Percentage points	95% CI	p	Percentage points	95% CI	p
Anxiety	Model 1	-0.32	(-0.80, 0.15)	0.1842	0.22	(-1.09, 1.54)	0.7399
	Model 2	-0.09	(-0.33, 0.14)	0.4318	0.15	(-1.04, 1.33)	0.8101
	Model 3	-0.11	(-0.49, 0.27)	0.5790	1.22	(-0.61, 3.04)	0.1970
Depression	Model 1	-0.05	(-0.30, 0.19)	0.6694	0.69	(-0.36, 1.74)	0.2020
	Model 2	0.11	(-0.05, 0.27)	0.1842	0.62	(-0.23, 1.48)	0.1597
	Model 3	-0.04	(-0.38, 0.30)	0.8165	1.47	(-0.41, 3.36)	0.1325
ADHD/ADD[d]	Model 1	-1.08	(-1.52, -0.65)	<0.0001***	-0.96	(-2.01, 0.08)	0.0773
	Model 2	-0.68	(-0.92, -0.43)	<0.0001***	-1.13	(-2.21, -0.05)	0.0454*
	Model 3	-0.66	(-1.01, -0.32)	0.0002***	-0.83	(-2.02, 0.37)	0.1818
Behavioral problems	Model 1	-0.72	(-1.03, -0.42)	<0.0001***	0.83	(-0.77, 2.43)	0.3127
	Model 2	-0.34	(-0.56, -0.12)	0.0021**	0.72	(-0.74, 2.18)	0.3413
	Model 3	-0.41	(-0.85, 0.03)	0.0708	2.51	(-0.49, 5.51)	0.1071

* <0.05, ** <0.01, *** <0.001

[a] Model 1 was unadjusted.

[a] Model 2 was adjusted for individual-level covariates including child age, sex, race and ethnicity, family nativity status, family structure, and highest adult education.

[a] Model 3 was additionally adjusted for state-level covariates including state minimum wage, Medicaid income eligibility limits for children aged 1-5 and 6-18, and EITC status.

[b] Low-income was defined as <200% of the Federal Poverty Level.

[c] All TWFE models included fixed effects for both state and year.

[d] Attention-deficit and hyperactivity disorder (ADHD/ADD).

HIGHLIGHTS

- Cash benefits varied between states and rose in 27 states and DC from 2015 to 2021
- Children in states with higher benefits had a lower probability of ADHD/ADD
- Within-state changes in benefit generosity were not linked to child mental health

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Supplementary Material
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